

# Applying Exponent Laws

## Lesson Plan

### 1. Lesson Information

|        |                                      |              |              |
|--------|--------------------------------------|--------------|--------------|
| Course | Grade 7 Accelerated<br>(1205050)     | Unit         | 3            |
| Lesson | Lesson 5 — Applying<br>Exponent Laws | Date         | _____        |
| Length | 53 minutes (one period)              | Benchmark(s) | MA.8.NSO.1.3 |

### 2. Standards

**MA.8.NSO.1.3** Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational-number bases, with procedural fluency.

**Must.** Rational-number bases with integer exponents: rewrite with positive exponents, and apply several laws to generate an equivalent expression.

**Must not.** No fractional exponents. Bases are nonzero, and the exponent's sign never moves the base's sign.

### 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR                | What it asks                                                                        | The evidence in this lesson                                                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MA.K12.MTR.<br>5.1 | Use patterns and structure to help understand and connect mathematical concepts.    | Notes I — 3.04's halving table redone with a fraction base, so the flip appears as the next line of a pattern the class already owns rather than a second rule. |
| MA.K12.MTR.<br>4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | Board 5 — when the room splits between $(-8/7)^2$ and $(8/7)^2$ , the sixty-second re-vote before the reveal.                                                   |
| MA.K12.MTR.<br>6.1 | Assess the reasonableness of solutions.                                             | Board 3 — a base between $-1$ and $0$ raised to a negative even power; students say which side of $1$ the value lands on before computing it.                   |

### 4. Learning Target and Essential Question

**Learning target.** I can rewrite a rational base with a negative exponent using a positive exponent, and apply several laws to generate an equivalent expression.

**Essential question.** How do you rewrite a rational base with a negative exponent as an equivalent expression with a positive exponent?

**Building on.** 3.04 — the negative exponent law; 3.02 — power of a quotient; Grade 6 — reciprocals of fractions.

**Working toward.** 3.06–3.07 — evaluating and testing equivalence with everything at once; Thread A — the same laws with variable bases.

## 5. Sequence

*read from the deck; never typed*

| Segment                             | Min       | What the teacher does                                                                        | What students do                           |
|-------------------------------------|-----------|----------------------------------------------------------------------------------------------|--------------------------------------------|
| Settle in; post the learning target | 1         | Post the target; say the 'today' line and nothing else yet.                                  | Copy the target.                           |
| Warm-Up                             | 5         | Two minutes silent, then reveal. One sentence of reteach per question at most.               | Four retrieval questions, alone, no notes. |
| Notes                               | 11        | Teach from the front. Expand once per law; say the OFF line out loud.                        | Copy the notes; answer the checks aloud.   |
| Example 1                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Example 2                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Whiteboards                         | 13        | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three.    |
| Independent Set                     | 6         | Silent. Circulate and mark what you see; do not teach.                                       | Six questions, written, alone.             |
| IXL                                 | 5         | Assign the day's IXL skills.                                                                 | Start the skills; finish tonight.          |
| <b>Total</b>                        | <b>53</b> |                                                                                              |                                            |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question                                                                                                                                              | DOK   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Marco says two negatives make a positive, so $(-2/3)^{-5}$ is $(2/3)^5$ . Which negative is he cancelling, and what does each of the two actually do? | DOK 3 |
| $4^{-3}$ and $(1/4)^3$ are the same number. Write 4 as a fraction and show why they have to be.                                                       | DOK 2 |
| For which bases is $(a/b)^{-n}$ larger than 1, and for which is it smaller? Give an example of each.                                                  | DOK 3 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks                                                                                                                | Answer                        | If the board is wrong                                                                                                                                                                                                                                                                                          |
|---|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | A recipe is scaled by a factor of $(2/3)^{-4}$ . The original recipe serves 6 people. Rewrite the factor with a positive exponent. | $\left(\frac{3}{2}\right)^4$  | $(2/3)^4$ — the minus was dropped without flipping [8.NSO.1.3 negative exponent law]; $-(3/2)^4$ — negative exponent read as a negative value [B1G-M misconception]; anything built from 6 — the unneeded figure was used                                                                                      |
| 2 | $\left(\frac{3}{-5}\right)^{-9}$                                                                                                   | $\left(\frac{-5}{3}\right)^9$ | $(5/3)^9$ — took the opposite as well as the reciprocal [B1G-M 8.NSO.1.3 misconception: opposite instead of reciprocal]                                                                                                                                                                                        |
| 3 | $\left(-\frac{8}{9}\right)^{-2}$                                                                                                   | $\frac{81}{64}$               | $-81/64$ — sign kept through an even power; $64/81$ — never flipped [8.NSO.1.3 negative exponent law]                                                                                                                                                                                                          |
| 4 | $\left(\frac{21}{4}\right)^{-5}$                                                                                                   | $\left(\frac{4}{21}\right)^5$ | $(21/4)^5$ — minus dropped without flipping; $4/21^5$ — flipped but applied the exponent to the denominator only                                                                                                                                                                                               |
| 5 | $\left(-\frac{7}{8}\right)^{-2}$ Which expression is equivalent?                                                                   | $A - (-8/7)^2$                | (B) took the opposite as well as the reciprocal [B1G-M 8.NSO.1.3 misconception: opposite instead of reciprocal]; (C) negative exponent read as a negative value [B1G-M 8.NSO.1.3 misconception: $-b$ vs $b^{-1}$ ]; (D) dropped the minus on the exponent without flipping [negative exponent law not applied] |
| 6 | $\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$ Which expression is equivalent?                                 | $A - (5/8)^2$                 | (B) flipped a base whose exponent was already positive [negative exponent law applied where no negative exponent remained]; (C) added $-2 + (-4)$ instead of subtracting [quotient law: subtract the exponents]; (D) multiplied $-2 \cdot -4$ [confused with power of a power]                                 |
| 7 | $\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1$                                                                   | $\frac{25}{4}$                | 1 — treated $(2/5)^{-1}$ as $2/5$ and multiplied by its reciprocal [negative exponent law not applied]; $5/2$ — only one factor counted                                                                                                                                                                        |
| 8 | $\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$                                                    | $\left(\frac{2}{3}\right)^9$  | $(2/3)^6$ — added $3 + 3$ in the power-of-a-power step [confused with product of powers]; $(3/2)^9$ — flipped a positive exponent                                                                                                                                                                              |
| 9 | Marco says $(-2/3)^{-5}$ can be rewritten as $(2/3)^5$ because 'two negatives make a positive.' Is he right?                       | $No - (-2/3)^{-5} = (-3/2)^5$ | 'Yes' — opposite taken as well as reciprocal [B1G-M 8.NSO.1.3 misconception: opposite instead of reciprocal]                                                                                                                                                                                                   |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / IEP.** Give the rule as one picture — the fraction with a curved arrow turning it over and the exponent's minus crossed out — and let the student redraw it at the top of the page. On the first four boards the only job is the flip; the arithmetic comes after.

**ELL.** Flip is the word. Say 'flip the fraction, drop the minus' on every board, and keep  $2/3 \rightarrow 3/2$  written up as the picture of what flip means.

**Enrichment.** Ask for  $0.4^{-2}$  by converting the decimal to a fraction first, and for the  $n$  that makes  $(2/3)^n = 81/16$ .

## 10. Closure, Vocabulary, Materials, Practice

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**Closure.** Board 9 is written work and it is the exit evidence: the student has to say separately what the base's sign does and what the exponent's sign does. Read the boards, not the papers.

**Vocabulary.** **reciprocal of a fraction; rational base with a negative exponent**

**Materials.** Whiteboards and markers. Calculators are allowed on everything;  $(-7/8)^{-2}$  on a calculator returns 1.306..., which is  $64/49$  — worth converting once so students trust both forms.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Identify equivalent expressions involving exponents I — VLM.